INTRODUCTION TO BUSINESS ANALYTICS

Student Project Roadmap & Requirements

Project Format: Each group will complete TWO Business Analytics projects using two different datasets and two different analytical tools.

Project 1 uses Power BI. Project 2 uses Microsoft Excel with Power Pivot, PivotTables, and PivotCharts.

Main Goal: Apply the Business Analytics process from understanding the business problem and cleaning data to analysis, dashboard creation, insights, and recommendations.

1. General Project Requirements

- Use a different dataset for each project.
- Choose datasets that contain enough records and variables for meaningful business analysis.
- Define a clear business problem before starting the analysis.
- Clean and prepare the data before building the analysis.
- Create a dashboard that answers specific business questions.
- End the project with business insights and practical recommendations.
- All calculations, transformations, and visuals must be created and understood by the students.

2. Project 1 — Power BI

Objective

Use Power BI to complete a full Business Analytics workflow, including Power Query data cleaning, data modeling , interactive visualizations, and a business dashboard.

Part 1 — Business Understanding

- Identify the company/industry represented by the dataset.
- Write a clear business problem.
- Write the main project objective.
- Identify the main stakeholders.
- Write at least 5 business questions.

Part 2 — Data Understanding

- Identify the dataset source.
- State the number of rows and columns.
- Describe the important variables/fields.
- Identify numerical, categorical, and date fields.

Part 3 — Data Cleaning & Transformation using Power Query

- Load the dataset into Power BI.
- Check and rename unclear column names.
- Check and correct data types.
- Identify and handle missing/null values.
- Identify and remove duplicate records where appropriate.
- Clean inconsistent text/category values.
- Remove unnecessary columns.

- Create or transform date fields where needed.
- Create useful derived/conditional columns when appropriate.
- Document the important cleaning and transformation steps.

Part 4 — Data Modeling

- Create relationships between tables if multiple tables are used.
- Check relationship direction and cardinality.

Part 5 — Power BI Dashboard (these are just suggestions each data has its own Dashboard)

- Create ONE professional interactive dashboard.
- Include KPI cards.
- Include at least 5 appropriate visuals.
- Include at least 2 interactive slicers.
- Use suitable chart types for the data.
- Include at least one trend/time analysis when a date field is available.
- Include category/product/customer/region analysis where applicable.
- Use clear titles and readable labels.
- Apply consistent formatting, alignment, and spacing.
- Do not overcrowd the dashboard.

Part 6 — Insights & Recommendations

- Write at least 5 business insights based on the dashboard.
- Write at least 3 business recommendations.
- Connect each recommendation to an analytical finding.
- Do not only describe a chart; explain what the result means for the business.

3. Project 2 — Excel + Power Pivot

Objective

Use Microsoft Excel to perform data cleaning, analysis, data modeling through Power Pivot, PivotTables/PivotCharts, and dashboard development.

Part 1 — Business Understanding

- Identify the company/industry represented by the dataset.
- Write a clear business problem.
- Write the main project objective.
- Identify the main stakeholders.
- Write at least 5 business questions.

Part 2 — Data Understanding

- Identify the dataset source.
- State the number of rows and columns.
- Describe the important variables.
- Identify numerical, categorical, and date fields.
- Write 3–5 initial observations.

Part 3 — Data Cleaning in Excel

- Open and inspect the raw dataset.
- Convert the data into an Excel Table.
- Check and correct data types.
- Identify and handle missing values.
- Identify and remove duplicates where appropriate.
- Clean inconsistent text/category values.
- Remove unnecessary columns.
- Create useful calculated/derived fields using Excel where appropriate.
- Format the dataset clearly and consistently.

Part 4 — Power Pivot & Data Model

- Load the cleaned Excel data into the Data Model/Power Pivot.
- Create relationships if multiple tables are used.
- Check the relationships and field names.
- Create useful Power Pivot measures.
- Create measures for the main business KPIs.
- Use the Data Model as the source for the analysis.

Part 5 — PivotTables & PivotCharts

- Create a PivotTable for the main KPI totals.
- Create a PivotTable for time/trend analysis.
- Create a PivotTable for category/product analysis.
- Create a PivotTable for customer/segment analysis when applicable.
- Create a PivotTable for geographic analysis when applicable.
- Create a Top/Bottom analysis where meaningful.
- Create PivotCharts from appropriate PivotTables.

Part 6 — Excel Dashboard (these are just suggestions each data has its own Dashboard)

- Create ONE professional dashboard using Excel, PivotTables/PivotCharts, and interactive controls.
- Include at least 2 dynamic KPI cards.
- Include at least 5 appropriate charts/visuals.
- Use PivotCharts where appropriate.
- Include at least 2 slicers.
- Connect slicers to all relevant PivotTables.
- Include a time/trend visual when a date field is available.
- Include category/product/customer/region analysis when applicable.
- Use clear dashboard titles and labels.
- Hide unnecessary gridlines and format the dashboard professionally.
- Use consistent colors, fonts, alignment, and spacing.
- Test that slicers and dashboard elements update correctly.

Part 7 — Insights & Recommendations

- Write at least 5 business insights.
- Write at least 3 business recommendations.
- Connect every recommendation to an analytical finding.

4. Required Business Questions

Each group must create questions that can be answered from the dataset. Questions should lead to analysis, not simple data lookup.

- Which product/category generates the highest sales?
- Which product/category generates the highest profit?
- How does performance change over time?
- Which region/location performs best and worst?
- Which customers or segments contribute most to the business?
- What factors or categories appear to be associated with higher profitability?
- What area should management focus on improving?

5. Dashboard Minimum Requirements (these are just suggestions each data has its own Dashboard)

Requirement	Power BI	Excel
KPI Cards	Minimum 4	Minimum 2
Visuals	Minimum 5	Minimum 5
Interactive Filters	Minimum 2 slicers	Minimum 2 slicers
Time Analysis	When date is available	When date is available
Dashboard	1 interactive dashboard	1 interactive dashboard
Insights	Minimum 5	Minimum 5
Recommendations	Minimum 3	Minimum 3

6. Final Deliverables

Project 1 — Power BI

- Power BI .pbix file
- Original dataset
- Project report/document
- Dashboard screenshots if requested
- Final presentation

Project 2 — Excel

- Excel .xlsx file with Power Pivot/Data Model and dashboard
- Original dataset
- Project report/document
- Dashboard screenshots if requested
- Final presentation

7. Project Report Structure

1. Cover Page

2. Business Problem

3. Project Objectives

4. Business Questions

5. Dataset Description

6. Data Cleaning & Preparation

7. Data Modeling / Analysis

8. Dashboard

9. Business Insights

10. Recommendations

11. Conclusion

12. References / Data Source

8. Final Presentation

- Presentation time: 15 minutes per group.
- Explain the business problem and objective.
- Briefly explain the dataset and important cleaning steps.
- Show the final dashboard.
- Answer the main business questions.
- Present the most important insights.
- Present the recommendations.
- Focus on the business value of the analysis, not only software steps.